

Comprehensive Literature Review

Finance-Focused NLP Chatbot and Algorithmic Trading Project

1 NLP for Finance Papers

1.1 A Survey on Large Language Models for Critical Societal Domains: Finance, Healthcare, and Law (2024)

Authors: Zhiyu Zoey Chen, Jing Ma, Xinlu Zhang, Nan Hao, An Yan, et al.

Publication: <https://arxiv.org/abs/2405.01769>

1.1.1 Summary

This comprehensive survey explores how large language models (LLMs) like GPT-3 and GPT-4 are transforming the financial sector alongside healthcare and law. The paper examines the methodologies, applications, challenges, and opportunities of LLMs within these high-stakes domains. In finance specifically, the authors highlight how LLMs are enhancing financial analytics, improving decision-making processes, and revolutionizing customer interactions. The survey also addresses ethical concerns and regulatory compliance issues that arise when deploying LLMs in finance.

1.1.2 Alignment with Our Project

This survey directly aligns with our project's core focus on developing a finance-focused NLP chatbot. It provides a broad overview of the current state of LLMs in finance, which serves as an excellent foundation for our work. The paper's discussion of challenges in applying LLMs to finance—such as specialized terminology, data privacy concerns, and regulatory requirements—helps us anticipate potential obstacles in our own implementation.

1.1.3 Key Learnings and Applications

- The survey identifies retrieval-augmented generation (RAG) as a promising approach for financial applications, which supports our methodology choice
- It emphasizes the importance of domain adaptation for financial LLMs, confirming our plan to fine-tune models on financial texts
- The paper highlights the need for transparent and explainable AI systems in finance, which we should incorporate into our chatbot design

- The authors discuss the value of combining structured financial data with unstructured text, supporting our integrated approach

1.1.4 Further Research Ideas

- Explore specialized pre-training techniques for financial domain adaptation mentioned in the survey
- Investigate the regulatory frameworks discussed to ensure our chatbot complies with financial industry standards
- Consider implementing the explainability techniques mentioned to make our chatbot's recommendations more transparent
- Research the evaluation frameworks suggested for assessing financial LLMs' performance

1.2 Finance is a Jungle and Text Embeddings Must Adapt (2024)

Authors: Peter Anderson, Mano Vikash Janardhanan, Jason He, Wei Cheng, Charlie Flanagan

Publication: EMNLP 2024. <https://arxiv.org/abs/2411.07142>

1.2.1 Summary

This paper addresses the challenge of specialized financial terminology in NLP applications. The authors present BAM embeddings, a set of text embeddings specifically optimized for financial document retrieval. These embeddings are trained on a carefully curated dataset of 14.3M query-passage pairs from both public and proprietary financial documents. The research demonstrates that domain-specific embeddings significantly outperform general-purpose embeddings on financial tasks, achieving 62.8% Recall@1 on financial queries (compared to 34.3% for multilingual-E5) and increasing question answering accuracy by 8% on FinanceBench.

1.2.2 Alignment with Our Project

The paper directly addresses one of the key challenges in our project: effectively processing and retrieving financial information. Our chatbot needs to understand specialized financial terminology and accurately retrieve relevant information, making these domain-specific embeddings highly relevant to our work. The improved performance on question answering tasks is particularly important for our chatbot's ability to interpret and respond to user queries about financial statements and investment opportunities.

1.2.3 Key Learnings and Applications

- We should consider using financial domain-specific embeddings like BAM instead of general-purpose embeddings for our retrieval system
- The paper provides insights into the importance of hard negative mining and data scale for training effective embeddings
- The authors' approach to evaluating embeddings on financial tasks provides a useful framework for assessing our own retrieval system
- Their finding that vector search with specialized embeddings outperforms lexical search for longer, more detailed queries is valuable for our query processing pipeline

1.2.4 Further Research Ideas

- Explore the possibility of fine-tuning existing embedding models on financial data using the techniques described
- Investigate combining lexical search with vector search as suggested in the paper for optimal performance across query types
- Consider creating a specialized evaluation dataset for our specific financial use cases, following their methodology
- Research how to effectively deploy these embeddings in a production environment alongside our LLM

1.3 BizBench: A Quantitative Reasoning Benchmark for Business and Finance (2024)

Authors: Koncel-Kedziorski et al.

Publication: <https://arxiv.org/abs/2311.06602>

1.3.1 Summary

This paper introduces BizBench, a comprehensive benchmark designed to evaluate language models on quantitative reasoning tasks in business and finance. The benchmark consists of eight distinct tasks that require models to perform numerical reasoning on both structured and unstructured financial data. These tasks include financial statement analysis, ratio calculation, valuation modeling, and other quantitative challenges that financial professionals regularly encounter. The authors evaluate several state-of-the-art language models on BizBench, revealing significant performance gaps compared to human experts.

1.3.2 Alignment with Our Project

Quantitative reasoning is a critical capability for our finance-focused chatbot, as users will likely ask questions requiring numerical analysis of financial statements and market data. BizBench provides a standardized way to evaluate our chatbot's performance on these tasks and identify areas for improvement. The benchmark's focus on real-world financial reasoning tasks makes it directly relevant to our project's goals of providing valuable financial insights and investment recommendations.

1.3.3 Key Learnings and Applications

- The benchmark identifies specific areas where LLMs struggle with financial reasoning, helping us prioritize improvements
- The paper provides examples of financial reasoning tasks that we should ensure our chatbot can handle
- The evaluation methodology offers a framework for assessing our chatbot's quantitative capabilities
- The authors' analysis of model performance across different task types can guide our approach to handling various financial queries

1.3.4 Further Research Ideas

- Use BizBench as a continuous evaluation tool during our chatbot development to track improvements
- Develop specialized modules for the specific reasoning tasks identified in the benchmark
- Explore techniques for enhancing LLMs' numerical reasoning capabilities in financial contexts
- Consider creating synthetic training data that targets the challenging aspects of financial reasoning identified in the paper

1.4 A Strategic Framework for Financial LLM Integration (2025)

Publication: <https://www.arxiv.org/abs/2504.02165>

1.4.1 Summary

This paper presents a practical framework for integrating LLMs into financial institutions. It addresses the unique challenges of implementing AI in financial services, including regulatory compliance, data security, and the need for accuracy in high-stakes decisions. The framework walks practitioners through evaluating whether advanced language capabilities are warranted, formalizing data governance, establishing appropriate use cases, and implementing effective oversight mechanisms. The authors provide case studies of successful LLM integration in credit assessment, personalized client services, and financial research automation.

1.4.2 Alignment with Our Project

While our project focuses on developing a new system rather than integrating LLMs into an existing financial institution, this paper provides valuable insights into practical considerations for financial AI applications. The framework's emphasis on appropriate use cases, data governance, and oversight aligns with our need to create a responsible and effective financial chatbot. The case studies offer examples of successful financial LLM applications that can inspire our own implementation.

1.4.3 Key Learnings and Applications

- The paper's discussion of appropriate use cases helps us refine our chatbot's functionality to focus on high-value applications
- The framework for data governance provides guidance for handling sensitive financial information
- The authors' approach to model selection and evaluation offers practical considerations for our own model choices
- The implementation strategies discussed can inform our development process and deployment planning

1.4.4 Further Research Ideas

- Adapt the governance framework for our specific application to ensure responsible AI use
- Explore the specific use cases mentioned in depth to understand best practices and potential pitfalls
- Investigate the oversight mechanisms discussed to develop appropriate monitoring for our chatbot

- Consider how the integration challenges identified might apply to our system's deployment in real-world settings

1.5 FinQA: A Dataset of Numerical Reasoning over Financial Data (2021)

Authors: Zhiyu Chen, Wenhui Chen, Charese Smiley, Sameena Shah, Iana Borova, Dylan Langdon, Reema Moussa, Matt Beane, Ting-Hao Huang, Bryan Routledge, William Yang Wang

Publication: EMNLP 2021. <https://arxiv.org/abs/2109.00122>

1.5.1 Summary

This paper introduces FinQA, a dataset specifically designed for numerical reasoning over financial data. The dataset consists of question-answer pairs that require models to perform complex calculations and reasoning based on financial reports. The questions are designed to mimic real-world financial analysis tasks, requiring operations such as addition, subtraction, multiplication, division, and more complex financial calculations. The authors also propose a program-based framework that decomposes numerical reasoning into executable programs, achieving better performance than traditional methods.

1.5.2 Alignment with Our Project

Numerical reasoning is a critical capability for our finance-focused chatbot, as users will frequently ask questions that require calculations based on financial statements and market data. FinQA provides both a valuable training resource and an evaluation benchmark for this specific capability. The program-based approach to numerical reasoning offers a potential methodology for implementing this functionality in our chatbot, ensuring accurate and transparent calculations.

1.5.3 Key Learnings and Applications

- The dataset provides realistic examples of financial questions requiring numerical reasoning, which can guide our training data creation
- The program-based framework offers a potential approach for implementing numerical reasoning in our chatbot
- The paper identifies common challenges in financial numerical reasoning, helping us anticipate and address these issues
- The evaluation methodology provides a framework for assessing our chatbot's numerical reasoning capabilities

1.5.4 Further Research Ideas

- Adapt the program-based framework for our specific use case, potentially extending it to handle more complex financial calculations
- Explore combining the program-based approach with neural methods for more flexible reasoning
- Investigate how to explain the reasoning process to users, making the calculations transparent and trustworthy
- Consider extending the dataset with additional question types relevant to our specific application domains

1.6 FinBERT: Financial Sentiment Analysis with Pre-trained Language Models (2020)

Authors: Dogu Tan Araci

Publication: Journal of Finance and Data Science. <https://arxiv.org/abs/1908.10063>

1.6.1 Summary

This paper introduces FinBERT, a domain-specific language model pre-trained on financial communication text and fine-tuned for financial sentiment analysis. The author adapts BERT to the financial domain by continuing pre-training on a large corpus of financial text from Reuters and SEC filings. The model is then fine-tuned on a financial sentiment analysis dataset. FinBERT significantly outperforms general-purpose language models on financial sentiment classification tasks, demonstrating the value of domain adaptation for financial NLP applications.

1.6.2 Alignment with Our Project

Sentiment analysis of financial news and reports is a key component of our project, as it will inform both the chatbot's responses and potentially the algorithmic trading decisions. FinBERT provides a proven approach to financial sentiment analysis that we can incorporate into our system. The paper's demonstration of the benefits of domain adaptation aligns with our plan to use finance-specific models for improved performance.

1.6.3 Key Learnings and Applications

- The paper demonstrates the effectiveness of domain adaptation for financial NLP tasks, supporting our approach

- FinBERT's architecture and training methodology provide a blueprint for our own domain adaptation efforts
- The sentiment analysis capabilities can be directly incorporated into our pipeline for analyzing financial news
- The evaluation methodology offers insights for assessing our own sentiment analysis component

1.6.4 Further Research Ideas

- Explore extending FinBERT's approach to more recent model architectures like RoBERTa or DeBERTa
- Investigate fine-tuning FinBERT on additional financial NLP tasks beyond sentiment analysis
- Consider how to effectively combine sentiment signals from multiple sources for more robust analysis
- Research methods for adapting the sentiment analysis to specific financial sectors or asset classes

2 Algorithmic Trading with Reinforcement Learning Papers

2.1 Adaptive Algorithm for Selecting the Optimal Trading Strategy Based on Reinforcement Learning for Managing a Hedge Fund (2024)

Authors: B. Belyakov, D. Sizykh

Publication: IEEE Access, Volume 12. <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=arnumber=107>

2.1.1 Summary

This paper presents a novel approach to hedge fund management using reinforcement learning, specifically the Proximal Policy Optimization (PPO) algorithm. The authors develop a framework that dynamically selects optimal trading strategies based on changing market conditions. The model considers diverse strategies including Mean Reversion and Momentum, and employs advanced mathematical frameworks to evaluate and select the most appropriate strategy at any given time. The research demonstrates superior performance compared to traditional methods, with improved Sharpe ratio, CVaR, and CDaR metrics. The authors emphasize that their approach is not a universal forecasting method but rather an advanced framework for trading fundamental strategies through RL.

2.1.2 Alignment with Our Project

This research directly aligns with our project's algorithmic trading component, particularly our focus on momentum and mean reversion strategies using reinforcement learning. The paper's approach to strategy selection provides a valuable framework for our implementation, offering insights into how to dynamically adapt trading strategies based on market conditions. The use of PPO, which we also plan to implement, provides specific guidance on algorithm selection and implementation.

2.1.3 Key Learnings and Applications

- The paper demonstrates the effectiveness of PPO for trading strategy selection, validating our algorithm choice
- The authors' approach to strategy evaluation using financial metrics provides a framework for our performance assessment
- The research highlights the importance of adapting to changing market conditions, supporting our hybrid strategy approach
- The implementation details offer practical guidance for our own reinforcement learning implementation

2.1.4 Further Research Ideas

- Explore the specific mathematical frameworks used for strategy evaluation and adapt them for our use case
- Investigate the authors' approach to handling non-stationary market data, a key challenge in financial RL
- Consider implementing their method for decomposing PnL structure to better understand trading performance
- Research how to scale their approach to a broader range of assets and market conditions

2.2 Deep Reinforcement Learning with Positional Context for Intraday Trading (2024)

Authors: Sven Goluža, Tomislav Kovačević, Tessa Bauman, Zvonko Kostanjčar

Publication: <https://arxiv.org/abs/2406.08013>

2.2.1 Summary

This paper explores the application of deep reinforcement learning for intraday trading, with a specific focus on incorporating positional context into the trading decision process. The authors develop a novel approach that enhances traditional RL methods by explicitly modeling the trader's current position and market context. The research discusses both momentum strategies (which expect prices to maintain historical trends) and mean-reversion strategies (which expect prices to revert to their long-term mean), analyzing their effectiveness in different market conditions. The authors demonstrate that their position-aware RL approach outperforms traditional methods, particularly in volatile market conditions.

2.2.2 Alignment with Our Project

This research directly addresses our project's focus on implementing momentum and mean reversion strategies using reinforcement learning. The paper's discussion of these strategies' effectiveness in different market conditions provides valuable insights for our hybrid approach. The position-aware RL methodology offers a potential enhancement to our planned implementation, potentially improving performance by incorporating additional context into the decision-making process.

2.2.3 Key Learnings and Applications

- The paper provides a detailed analysis of when momentum vs. mean reversion strategies are most effective, informing our hybrid approach
- The position-aware RL methodology offers a potential enhancement to our implementation
- The authors' approach to state representation provides guidance for our own feature engineering
- The evaluation methodology offers a framework for assessing our trading strategies' performance

2.2.4 Further Research Ideas

- Implement the position-aware RL methodology described in the paper and compare it to traditional approaches
- Explore the authors' approach to identifying market regimes suitable for different strategies

- Investigate their method for handling transaction costs and market impact in the RL framework
- Consider extending their approach to incorporate additional market context from our NLP component

2.3 Pro Trader RL: Reinforcement learning framework for generating trading knowledge by mimicking the decision-making patterns of professional traders (2024)

Authors: Da Woon Jeong, Yeong Hyeon Gu

Publication: <https://dl.acm.org/doi/10.1016/j.eswa.2024.124465>.

link: <https://www.sciencedirect.com/science/article/pii/S0957417424013319?via>

2.3.1 Summary

This paper introduces Pro Trader RL, a novel reinforcement learning framework designed to mimic the decision-making patterns and trading philosophy of professional traders. Unlike traditional RL approaches that simplify trading into a single model, Pro Trader RL meticulously replicates the complex decision-making process of professional traders through four main modules: Data Preprocessing, Buy Knowledge RL, Sell Knowledge RL, and Stop Loss Rule. The framework takes into account the complex nature of the stock market and presents an integrated approach to RL, from data pre-processing to buying, selling, and implementing stop-loss mechanisms. Experimental results demonstrate that the framework achieves high returns and Sharpe ratio regardless of market conditions and maintains stable performance with low maximum drawdown.

2.3.2 Alignment with Our Project

This research offers valuable insights for our algorithmic trading component, particularly in how to structure the reinforcement learning approach to reflect professional trading practices. The modular design of Pro Trader RL aligns with our need to develop a comprehensive trading system that handles different aspects of the trading process. The framework's emphasis on risk management through stop-loss mechanisms is particularly relevant to our secondary objective of ensuring practical usability with appropriate risk management features.

2.3.3 Key Learnings and Applications

- The modular approach to RL for trading provides a potential structure for our own implementation

- The separation of buy and sell decisions into distinct modules offers a more nuanced approach than traditional methods
- The integration of stop-loss rules directly into the framework provides guidance for our risk management implementation
- The authors' approach to reward calculation offers insights for designing our own reward functions

2.3.4 Further Research Ideas

- Adapt the modular structure of Pro Trader RL for our specific trading strategies
- Explore the authors' approach to relative reward calculation for more precise reinforcement signals
- Investigate their method for integrating risk management directly into the RL framework
- Consider how to incorporate insights from our NLP component into this type of modular trading framework

2.4 Combining mean reversion and momentum trading strategies in foreign exchange markets (2024)

Publication: Research Gate. https://www.researchgate.net/publication/222554542Combining_mean_reversion_and_momentum_trading_strategies_in_foreign_exchange_markets

2.4.1 Summary

This paper implements a trading strategy that combines mean reversion and momentum approaches in foreign exchange markets. Originally designed for equity markets, the authors adapt the strategy for forex trading, addressing the unique characteristics of currency markets. The research explores how these seemingly contradictory strategies can be effectively combined to capture different market behaviors at different time scales or under different conditions. The authors develop a framework for determining when to apply each strategy and how to weight their signals, demonstrating that the combined approach outperforms either strategy used in isolation across various currency pairs and market conditions.

2.4.2 Alignment with Our Project

This research directly addresses our project's focus on implementing and combining momentum and mean reversion strategies. While the paper focuses on forex markets rather than equities, the principles of strategy combination are applicable to our work. The

framework for determining when to apply each strategy provides valuable insights for our hybrid approach, potentially improving performance by leveraging the strengths of both strategies while mitigating their individual weaknesses.

2.4.3 Key Learnings and Applications

- The paper provides a practical framework for combining momentum and mean reversion strategies, informing our hybrid approach
- The authors' method for determining market conditions suitable for each strategy offers guidance for our implementation
- The research demonstrates the potential performance improvements from a combined approach, validating our strategy
- The evaluation across different market conditions provides insights into the robustness of the combined approach

2.4.4 Further Research Ideas

- Adapt the strategy combination framework for equity markets and our specific trading objectives
- Explore the authors' approach to signal weighting and strategy switching for our hybrid implementation
- Investigate how to incorporate NLP-derived market sentiment into the strategy selection process
- Consider extending their approach to include additional strategies beyond momentum and mean reversion

2.5 A Multifaceted Approach to Stock Market Trading Using Reinforcement Learning (2024)

Publication: IEEE. <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=arnumber=10570190>

2.5.1 Summary

This research presents a multifaceted strategy for multi-stock market trading using reinforcement learning. The authors develop an approach that incorporates enhanced state representation based on daily market data, including technical indicators, fundamental metrics, and market sentiment. The paper explores how different aspects of market data

can be effectively combined to create a more comprehensive trading strategy. The authors implement various RL algorithms, comparing their performance across different market conditions and asset classes. The research demonstrates that a multifaceted approach that considers diverse data sources outperforms simpler strategies, particularly in complex and volatile market environments.

2.5.2 Alignment with Our Project

This research aligns with our project's goal of developing a comprehensive trading approach that leverages multiple data sources. The paper's focus on enhanced state representation provides valuable insights for our feature engineering process. The multifaceted approach resonates with our plan to integrate NLP-derived insights with quantitative trading signals, creating a more holistic trading system. The comparison of different RL algorithms offers guidance for our algorithm selection and implementation.

2.5.3 Key Learnings and Applications

- The paper demonstrates the value of incorporating diverse data sources into the trading decision process, supporting our integrated approach
- The authors' method for state representation provides guidance for our feature engineering
- The comparison of different RL algorithms offers insights for our algorithm selection
- The evaluation across different market conditions provides a framework for assessing our strategy's robustness

2.5.4 Further Research Ideas

- Explore the specific technical indicators and fundamental metrics used in the paper for our state representation
- Investigate their approach to handling multiple assets simultaneously for portfolio management
- Adapt their multifaceted approach to incorporate our NLP-derived insights
- Consider implementing their methods for evaluating strategy performance across different market regimes

2.6 Deep Reinforcement Learning for Automated Stock Trading: An Ensemble Strategy (2022)

Authors: Hongyang Yang, Xiao-Yang Liu, Shan Zhong, Anwar Walid

Publication: ACM SIGKDD Conference on Knowledge Discovery and Data Mining.

<https://openfin.engineering.columbia.edu/sites/default/files/content/publications/ensemble.pdf>

2.6.1 Summary

This paper presents an innovative ensemble strategy for automated stock trading using deep reinforcement learning. The authors develop a framework that combines multiple RL algorithms, including Proximal Policy Optimization (PPO), Deep Deterministic Policy Gradient (DDPG), and Advantage Actor-Critic (A2C), to create a more robust trading system. Each algorithm has its strengths in different market conditions, and the ensemble approach leverages these complementary capabilities. The research demonstrates that the ensemble strategy significantly outperforms individual algorithms and traditional trading methods in terms of returns, Sharpe ratio, and maximum drawdown across various market conditions.

2.6.2 Alignment with Our Project

This research provides valuable insights for our algorithmic trading component, particularly in how to combine multiple RL approaches for improved performance. While our project focuses specifically on momentum and mean reversion strategies, the ensemble methodology could be adapted to combine these approaches effectively. The paper's comprehensive evaluation across different market conditions aligns with our need to develop robust trading strategies that perform well in various scenarios.

2.6.3 Key Learnings and Applications

- The ensemble approach offers a potential framework for combining different trading strategies or RL algorithms
- The comparison of PPO, DDPG, and A2C provides insights for our algorithm selection
- The authors' approach to portfolio management and risk control offers guidance for our implementation
- The evaluation methodology provides a comprehensive framework for assessing trading performance

2.6.4 Further Research Ideas

- Explore adapting the ensemble methodology to combine momentum and mean reversion strategies
- Investigate the specific strengths and weaknesses of different RL algorithms in various market conditions
- Consider implementing a dynamic weighting mechanism for the ensemble based on market conditions
- Research how to incorporate NLP-derived insights into the ensemble decision-making process

2.7 Reinforcement Learning for Market Making in a Multi-agent Dealer Market (2023)

Authors: Svitlana Vyetrenko, Shaojie Xu, Alberto Bietti, Tucker Balch, Xiao-Yang Liu

Publication: International Conference on Machine Learning (ICML)

2.7.1 Summary

This paper explores the application of reinforcement learning to market making in a multi-agent dealer market. Market making involves providing liquidity by continuously quoting buy and sell prices, profiting from the spread while managing inventory risk. The authors develop a multi-agent RL framework where agents learn to optimize their quoting strategies in a competitive environment. The research addresses key challenges in market making, including adverse selection, inventory management, and competition among dealers. The authors demonstrate that their RL approach outperforms traditional market making strategies, particularly in volatile and uncertain market conditions.

2.7.2 Alignment with Our Project

While our project doesn't specifically focus on market making, this research provides valuable insights into applying RL to complex financial decision-making problems. The paper's approach to managing risk and optimizing in a competitive environment is relevant to our algorithmic trading component. The multi-agent aspect could inform how our system adapts to markets where multiple algorithmic traders are operating simultaneously, which is increasingly common in modern financial markets.

2.7.3 Key Learnings and Applications

- The paper's approach to risk management and inventory control offers insights for our trading strategies

- The multi-agent framework provides perspective on how to adapt to competitive market environments
- The authors' state representation and reward design offer guidance for our RL implementation
- The evaluation in realistic market conditions provides benchmarks for assessing trading performance

2.7.4 Further Research Ideas

- Explore incorporating aspects of market making into our trading strategies for improved liquidity management
- Investigate the multi-agent training methodology for more robust strategy development
- Consider how to model and adapt to the behavior of other market participants
- Research how NLP-derived insights could provide advantages in competitive trading environments

3 Synthesis and Research Gap

Our comprehensive literature review reveals significant advancements in both NLP for finance and algorithmic trading with reinforcement learning. Please note that the papers highlighted here are a subset of our numerous literature reviews that we want to emphasize. The NLP papers demonstrate the potential of domain-specific models and embeddings for financial applications, while highlighting the challenges of specialized terminology and numerical reasoning. The algorithmic trading papers provide frameworks for implementing momentum and mean reversion strategies using reinforcement learning, with recent work exploring hybrid approaches, position-aware methods, and explainable techniques.

However, there remains a notable research gap in integrating these two domains. Most existing work treats NLP-based financial analysis and algorithmic trading as separate fields, with limited exploration of how insights from one can inform the other. Our project aims to bridge this gap by developing a system that combines qualitative insights from NLP-based analysis of financial news and reports with quantitative trading strategies.

The literature also highlights several key considerations for our project:

1. **Domain Adaptation:** Financial applications require specialized models and embeddings that understand domain-specific terminology and concepts. Papers like

FinBERT and BAM embeddings demonstrate the significant performance improvements achieved through domain adaptation.

2. **Numerical Reasoning:** Effective financial analysis requires the ability to perform complex calculations and reasoning based on financial data. FinQA and BizBench provide valuable resources for developing and evaluating these capabilities.
3. **Strategy Combination:** Hybrid approaches that combine momentum and mean reversion strategies can outperform single-strategy implementations. The research on combining these strategies in forex markets provides a valuable framework for our approach.
4. **Risk Management:** Incorporating risk management directly into trading frameworks is essential for practical applications. The Pro Trader RL framework and the explainable RL research offer insights into effective risk management approaches.
5. **Explainability:** Transparent and interpretable models are crucial for financial applications. The research on explainable RL for portfolio management demonstrates that explainability can be achieved without sacrificing performance.
6. **Multimodal Integration:** Combining textual and numerical data can provide more comprehensive financial insights. FLANG's approach to multimodal financial forecasting offers a potential framework for our integration efforts.

By addressing these considerations and bridging the gap between NLP and algorithmic trading, our project has the potential to make significant contributions to both academic research and practical applications in finance. The integration of qualitative insights from financial texts with quantitative trading strategies represents a novel approach that could enhance both the accuracy of financial analysis and the performance of trading decisions.